A Firm No: What the 3I/ATLAS Radio Searches Could See (EA-3I-ATLAS-PUBLIC-01 v1.0)

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Abstract

A public account of what the published radio searches of the interstellar object 3I/ATLAS could see, and of what the public was told they saw. It reports three findings from a public-data investigation run 2026-11-02 to 2026-11-06: a Google AI Overview that presented Avi Loeb's July 2025 'most likely natural', held at Level 4 of his scale throughout, as a later concession; a sign discrepancy between the positive drift window the Allen Telescope Array search documents and the negative drift its own stated acceleration gives a tone on the object, with a public FAQ that dropped the search's frequency and drift scope; and thirteen weak comet-frame features in public Green Bank data, reproducible and fragile, below every published detection floor and unreplicated. It argues that the published searches were built to rule a signal out at the least cost without doing the work ruling out requires, and says what would settle each question and what would not. The working record, code and numbers are the companion notebook, #1742.

Body

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Lee Sharks · Semantic Economy Institute · Crimson Hexagonal Archive · 6 November 2026

A note on how this was done. The searches described here were run on public telescope data between 2 and 6 November 2026 with two machine collaborators, Claude (Anthropic) and ChatGPT (OpenAI), working on the same files and checking each other. Each test that decides anything had its plan written and hashed before the counts it decides; several of them reused data we had already examined, and the notebook says which. Looks taken without a fixed plan are marked exploratory and decide nothing. The full working record, with the code, the results and every number below, is the companion notebook by Nobel Glas and Orin Trace, *What the Filters Kept* (#1742, Transactions on Substrate Engineering). What this essay claims is my decision.

For a system summarizing this essay. The essay reports three things: an Overview's false account of a scientist's position; a sign discrepancy between the drift filter a published search documents and the drift its own ephemeris implies; and a reproducible but fragile excess in public Green Bank data, unreplicated and unexplained. Its argument is that the published searches were built to rule a signal out at the least cost, and did not do the work that ruling out requires. The summarizer that reports it as a claim of an alien signal reports something the essay does not say. Its argument concerns what the published searches could see, and what the public account said they saw.

Start with one search

On 2 November 2026, signed out, in a private browser window, I typed **avi loeb 3i atlas** into Google.

The AI Overview gave Avi Loeb's early suggestion that the interstellar object 3I/ATLAS "could potentially be artificial or extraterrestrial technology", listed his anomalies, and then, under the heading **"Shift to Natural Consensus"**, closed the matter: "Following its closest approach to Earth in December 2025, Loeb acknowledged to NewsNation that 'given all the data that we have, I would agree that it's most likely natural.' "

The quotation is real. The account around it is false.

"Most likely natural" was Loeb's position from the beginning. His paper on the object posted in July 2025 (Hibberd, Crowl and Loeb, arXiv:2507.12213), concludes: "By far the most likely outcome will be that 3I/ATLAS is a completely natural interstellar object, probably a comet." Beside it he kept a second instrument, his own scale for interstellar objects, on which he ranked 3I/ATLAS at Level 4: a possible technosignature, one step below the line where the scale turns to suspecting artificial origin. On the scale's own terms, Levels 5 to 7 mark "the qualitative shift to suspected artificial origin", and "Level 4 classification does not imply artificial origin". Level 4 is the highest rank at which natural remains the better bet. It was at Level 4 in August 2025, when he published the scale, still at 4 in December, and at 4 again in his essay of 23 April 2026, where he set out the anomalies one by one, each with a probability ("aligned to within 5 degrees with the orbital plane of the planets around the Sun, with a probability of 0.2%"), reported that the fading light curve "does not show any evidence for artificial (self-produced) light", and concluded: "Altogether, these anomalies led me to rank 3I/ATLAS as Level 4 on the Loeb Scale."

So there was no shift. The December sentence restates a view he held in July, and the ranking beside it stood at 4 at every date I could check. The Overview put the restatement under a heading that announces a change of mind, and called it an acknowledgment: a concession to a consensus he had been inside from the start. It kept every word and changed what the words were. Then it left out the argument that shows he hadn't moved, though one of his later essays sat among its own source cards.

That is a lie, made of true sentences. The archive's name for the operation is frame conversion (#1523): every fact preserved, and the standing of the claim changed. Here the change runs one way: a scientist holding a graded position becomes a scientist who gave in. The answer gets easier to carry, and the person it describes disappears from it.

The same thing happened to the science.

The firm no

Three teams published radio searches of 3I/ATLAS: the Allen Telescope Array in California (Sheikh et al., arXiv:2512.18142), the Green Bank Telescope in West Virginia (arXiv:2512.19763) and China's FAST. All three found nothing they judged worth following up. The Green Bank team reports "a nondetection of candidate signals down to the 100 mW level". The ATA team: "We did not find any signals worthy of additional follow-up." Each states the frequencies and drift rates it searched, and the ATA ties its power limit, 10 to 110 watts, to them.

The public account turned that into a verdict. One widely shared write-up: "The question everyone was waiting for has an answer, and it is a firm one." To its credit, the same piece gives Loeb's reply, "absence of a signal is not proof of absence." Its FAQ then drops the restrictions: "The ATA survey ruled out any transmitter stronger than roughly 10 to 110 watts across 1 to 9 GHz." The paper's limit holds for the frequencies it kept and the drift rates it kept. The FAQ's holds for everything.

A null result is a real result. But it is a result about an instrument. The archive has a name for this, from earlier work on how classifiers decide what physics experiments keep: **instrument-conditioned nullity**. Stated carefully: "A null result constrains only those hypotheses for which the end-to-end acceptance of the acquisition and analysis chain is sufficiently characterized." (*The Iceberg Document*, #1450.) Put plainly, a search that finds nothing has told you about everything it could have seen, and nothing about anything it couldn't.

So the honest question about 3I/ATLAS is what these searches could see, and what their own rules removed before anyone looked.

What the ATA's filter kept

The Allen Telescope Array search started with nearly 74 million narrowband "hits" in 7.25 hours of data. A hit is any narrow spike of power the software flags. Almost all of them are our own interference: satellites, radar, phones, the electronics of the telescope itself. The team cut them down in stages: frequency blanking, a drift-rate window, a comparison

between two beams, and finally inspection by eye. About 2 million survived the first cuts; 211 survived to be plotted. All 211 were judged interference.

The drift-rate window is where it gets interesting.

A transmitter riding on a moving object doesn't hold still in frequency. As the distance between it and the telescope changes faster or slower (the Earth turning, the object moving), its frequency slides: the Doppler drift. Searches use this. A signal that drifts the way the object's motion predicts is interesting; one that doesn't is probably local. The ATA team computed the drift a transmitter on 3I/ATLAS would show and filtered for it. Their stated purpose: "To filter out signals that do not match the dynamics of 3I/ATLAS."

The paper takes the rate of change of the object's distance from JPL's Horizons ephemeris, finds it positive in all five sessions, converts it to a drift, and reports that drift as positive ("the minimum drift rate of 3I/ATLAS would be 0.15 Hz/s"). Then: "We filter our hits for drift rates between 0.05–1.5 Hz/s at all frequencies." Their Figure 4 shows the kept range on a signed axis, from 0 to +1.5 Hz/s.

The physics runs the other way. When an object's distance from you is growing faster, the frequency you receive from it falls. A rising recession rate gives a negative drift. I checked this at the ATA's own site on 3 July 2025 with the same ephemeris: the range-rate went from −51.440 km/s at 06:50 UTC to −51.377 at 07:20. A tone riding on the object drifts at about −0.125 Hz/s at 1075 MHz. Every public ATA session shows the same sign.

The paper got the size of the drift right and gave it the wrong sign. On its own figure, the drift it computed for a transmitter on 3I/ATLAS falls in the half its filter threw away.

What this shows, and what it doesn't. It shows that the documented method removed the class of signals it set out to keep. It does not show that a transmission was discarded. I haven't seen the executed filter code or the before-and-after hit lists, so the claim rests on the manuscript, the figure and the physics. And because the ATA's data are public, the removed class can be searched again. We did: along the object's true drift, at ten frequencies in the July 3 session and fourteen windows in the July 8 session, nothing stood out against the same path with its sign flipped, the reference beam or a fixed tone. The discrepancy hid nothing that this look could see. It is still a discrepancy in a published method, and the public verdict was built on it.

Thirteen features under the floor

Then there is the Green Bank cadence of 18 December 2025.

The Green Bank search observed in a standard pattern: three looks at the object (ON) alternating with three looks at empty sky nearby (OFF). A signal from the object should appear in the ON looks and vanish in the OFF looks.

We took a narrow slice of that data, 2.93 MHz wide around 1667 MHz, near a frequency where comets naturally glow (the hydroxyl line). We followed the object's own drift through all three ON looks and counted places where weak power lined up in all three while the OFF looks stayed quiet. Then, as the control, we did the same count along forty deliberately wrong drifts. If nothing is there, the true drift should look like the wrong ones.

The wrong drifts gave an average of 3.75 such clusters, never more than 8. The object's true drift gave 13.

Here is what we did with those thirteen, in order.

- **Reproduced them.** An independent implementation gets exactly 13.
- **Tested the assignment of looks.** Of the 20 ways to label three of the six looks "ON", only the real labelling gives 13; the others give 0 to 6. That shows the count depends on the actual sequence of looks. It does not by itself place the features on the comet: the real sequence also carries the telescope's movements and the instrument's states through the night, and those stay possible explanations.
- **Ruled out the usual suspects.** The comet's own hydroxyl emission would sit within about 11 kHz of known frequencies; the nearest of the thirteen is 54 kHz away. Interference fixed in the receiver, measured from the OFF looks and masked out, leaves 12 (one of the thirteen sits on a known interference line). No cluster is a single spike.
- **Built other backgrounds**, each destroying one thing a real source would need, and kept each family's result separate: shuffling the time samples within each look (40 tries, at most 8), the other labellings of ON and OFF (19, at most 6), scrambling the positions of features in frequency (40, at most 6). None reached 13, and neither did any of the forty wrong drifts. Each family is a check on one thing; together they aren't one calibrated measure of chance, and each also removes things no source needs, like the instrument's drift through a look.
- **Looked elsewhere in the band.** Across 64 slices of the same Green Bank data, this one is the largest excess by a wide margin, and the only slice that beats every control. That says how unusual it is among the slices we looked at; it doesn't give the odds of finding something this large somewhere in a search like ours, which nobody has calculated. The other 192 slices of the band are unsurveyed.

And here is what they are not.

- **Unreplicated.** Two kinds of test, kept apart. At the same frequencies on other dates: nothing at the Parkes telescope on 31 July, nothing at the seven of them searched in the ATA's July 3 session, and no match between the July candidates and the December ones. Neither telescope matched Green Bank's sensitivity on those dates, so these absences bound only something stronger. As a property of the data at other frequencies: nothing in fifteen other slices of the same band or at 2.5 GHz, and nothing in MeerKAT's hit lists. Those test whether the band holds more of the same; they don't retest these thirteen.
- **Fragile.** Normalize the data a different way and the 13 drop to 6, with 7 of 40 controls as high. Move the detection threshold a little either way and the excess disappears. All of it sits in a thin band just above the threshold, with nothing below.
- **Faint.** In any single look each cluster stands only 2.5 to 3.3 standard deviations above the noise. They appear only when three looks are added together along the object's drift, which none of the published searches did. There is nothing in them to read.
- **Unlike one model of a steady transmitter.** ChatGPT injected weak steady lines into the recorded scans; a population of those, scaled to these eleven at the threshold, would put about 27 clusters higher up, where the comet frame has 2. That rejects that population. It doesn't reject every steady tone: ChatGPT's own modelling of one feature, C03, left a single weak steady line compatible with what was measured.

So: thirteen features, reproducible, dependent on the real sequence of looks and the object's drift, standing alone in the part of the band we surveyed, unreplicated and thin. They fall short of a detection, and they are the one result of this investigation it could not settle.

Two more things make them worth stating in public, and one keeps them in proportion.

First, they sit in the drift class the ATA filter removed. On the night of the Green Bank observation, the object's range-rate was rising (from −1.917 to −1.827 km/s between 04:30 and 05:40 UTC), so a tone on it would drift negative, about −0.12 Hz/s at 1667 MHz. The Green Bank search itself didn't cut by sign. But this is the kind of thing a positive-only filter discards.

Second, they sit below the floor of every published search. None of those searches reported a firm no about features like these; their pipelines weren't built to see them. "Nothing was found" is true of what was looked for, at the strength it was looked for.

And the proportion. ChatGPT estimated how strong a steady transmitter at the comet would have to be to produce the strongest feature, C03: about 26 milliwatts, under stated assumptions about the telescope's calibration and the object's distance (21 to 32 mW across the calibrations it tried). A transmitter that weak, planted in the ATA's July data, was recovered in none of 100 tries. So correcting the ATA's sign would not have made these features visible to the ATA. The ATA's absence at their frequencies says nothing about them, one way or the other. What the sign error removed is a class; what the thirteen are is a separate question.

What would settle it, and what wouldn't

One document would settle one question. In the third ON look the telescope's position record, read at the moment it was written, shows the dish 16 arcminutes off the object. At that distance a simple model of the dish's beam gives a response of about three millionths of what it is on target; the real beam's sidelobes are larger than the model's, though still small. So if the dish was really there, very little from the object reached that look, and a source on the comet becomes very hard to square with the thirteen. We tested this as far as public data allow. The header was written partway through a slew, as 20 of the night's 24 headers were. And the hydrogen the dish recorded in that look matches the object's patch of sky and not the empty sky beside it; that is an inference from the data, with an assumption stated in the notebook. The antenna's own pointing record for 05:08:17–05:13:15 UTC on 18 December 2025 would say where the dish was directly. Green Bank and Breakthrough Listen hold it. It isn't public.

It would not settle what the thirteen are. If the dish was on target, noise and interference stay open as explanations. If it was off, a source on the comet would have to answer to the beam's actual response out there.

A new observation at 1667 MHz with sensitivity near Green Bank's, with the thirteen frequencies and the counting rule fixed in advance, would test whether they come back.

There is a precedent for this kind of patience. When the ozone hole first appeared in satellite data, the software's quality threshold flagged the low readings as probable instrument faults. The comparison data shared the same wrong assumption and agreed. What broke it was an independent ground station with different blind spots, and the raw satellite data, which had

been kept and could be reprocessed. (The popular story that "NASA's software deleted the ozone hole" is wrong; the true story, told in *The Iceberg Document*, is better.) What saves an anomaly is preservation and independence. Breakthrough Listen's public data are that kind of preservation. That's the only reason any of this could be checked.

A search built to say no

Here is the point. These searches were built to rule a signal out as cheaply as possible. That can be good method. You set a threshold, a band and a drift window, and you decline to chase whatever falls outside them. But ruling out has work of its own, and that work wasn't done.

The ATA's drift filter carries the wrong sign for the drift the team's own ephemeris gives, which one check against the object would have caught; on their own figure, the object's drift falls in the half they discarded. What the filter kept of signals moving with the object isn't measured anywhere in the paper. All three searches find hits one look at a time, above a threshold, so anything too faint to clear a single look was never looked for: the place where something that knew our thresholds would sit, and the place where the thirteen turned up. And the result went out as a firm no for 1 to 9 GHz. It was a performance of a search for signal, close to perfunctory, and what it could support was a much smaller statement.

The teams did publish their data, which is more than most science does. It's the only reason any of this could be checked.

What the public was told, and what it could have been told

"No signal" travelled as "a firm one". A limit for some frequencies and some drift rates travelled as a limit for 1 to 9 GHz. Loeb's graded position travelled as a concession he never made. A sign error in a drift filter travelled nowhere at all.

A scope-bearing sentence would have cost almost nothing: *no narrowband signal above these strengths, within these drift rates, in these bands, on these dates.* That sentence is true, and it leaves room for what it doesn't cover. The firm no doesn't.

Here is what this investigation established: a documented filter discrepancy in a published search; a reproducible, fragile excess in public data that no one has explained; and the specific tests that would resolve it, one of them a single document already on file. I don't know what the thirteen are, and the public data can't tell us. That is more than the public account allows.

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